

(a) Parallel scaling

$P=8$	2.036★	1.993★	1.932★	1.886★	1.822★	1.777★
$P=4$	2.054	2.013	1.951	1.904	1.839	1.794
$P=2$	2.077	2.036	1.973	1.927	1.861	1.814
$P=1$	2.111	2.067	2.003	1.954	1.888	1.845
	0.5B	0.7B	1.1B	1.6B	2.8B	4.4B

Pile: parameters N

★ SimpleTES closer (5/12) ★ EvoDuet closer (7/12)

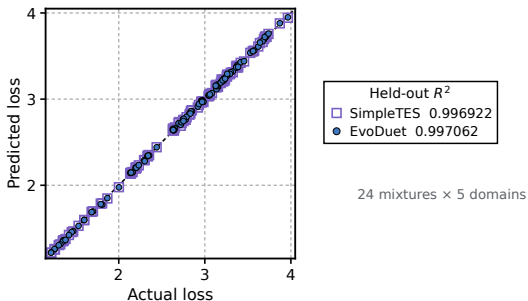
1.123★	1.100★	1.069★	1.038★	1.002★	0.979★
1.135	1.112	1.081	1.049	1.013	0.991
1.151	1.126	1.094	1.062	1.024	1.002
1.172	1.150	1.113	1.082	1.045	1.021
0.5B	0.7B	1.1B	1.6B	2.8B	4.4B

Stack: parameters N

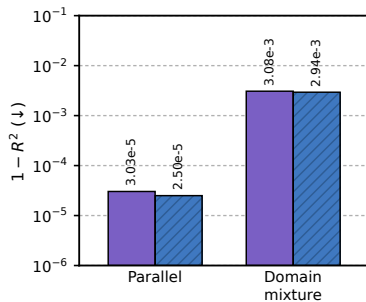
SimpleTES $L = (a + bN^{-c})(1 + dP^{-1/2})$
 EvoDuet $L = \beta_0 + \beta_1 u + \beta_2 v + \beta_3 uv$; $u = (N/10^9)^{-0.2}$, $v = (1 + 0.4 \log_2 P)^{-0.2}$

Held-out R^2
 SimpleTES 0.999970
 EvoDuet 0.999975

(b) Domain mixture



(c) Held-out error



SimpleTES $L = \mathbf{a} + \exp(W(\mathbf{d} \odot \mathbf{p}^c) + \mathbf{b})$
 EvoDuet $L_j = a_j + \exp(b_j + \mathbf{w}_j^T \mathbf{h}(\mathbf{p})) - c_j \ln(5p_j)$